

DATASET CARD FOR SENSE-7

OVERVIEW

SENSE-7 is a dataset designed to measure how real users perceive empathy in AI chatbots during multi-turn conversations. Instead of external annotators, the people actually chatting with the AI rate each of its responses. It was developed to capture user perceptions of empathy during multi-turn human–AI conversational interactions, focusing on subjective annotations from the human participant rather than third-party judgments.

See related <https://github.com/microsoft/sense-7>

A detailed discussion of SENSE-7, including how it was developed and evaluated, can be found in our paper at: <https://arxiv.org/abs/2509.16437>

INTENDED USES

SENSE-7 is best suited for studying and modeling how users perceive empathy in AI conversations. It enables research on empathy detection, prediction, and generation across multi-turn dialogues, helping to design more emotionally intelligent and user-sensitive AI systems and exploring how perceived empathy evolves during interactions.

SENSE-7 is being shared with the research community to facilitate reproduction of our results and foster further research in this area.

OUT-OF-SCOPE USES

SENSE-7 is not well suited for training large-scale, general-purpose, or open-domain dialog models, since it is relatively small from information workers. It is not ideal for clinical or therapeutic research, as participants were not patients or clients. It is not suited for cross-cultural, multi-lingual, or multimodal studies; only covers text-based conversations, so it's not suited for multimodal empathy research.

There are no instances of cultural and occupational diversity, non-English language use, multimodal empathy cues, and highly emotional or sensitive domains in this dataset. As a result, SENSE-7 is not expected to perform optimally at general-purpose, clinical/therapeutic, or cross-cultural empathic dialog modeling.

We do not recommend using SENSE-7 in commercial or real-world applications without further testing and development. It is being released for research purposes.

We do not recommend using SENSE-7 in the context of high-risk decision making (e.g. in law enforcement, legal, finance, or healthcare).

The dataset does not include information about potential risks or harm from interactions. As empathic interactions can increase trust and engagement, we do not recommend using SENSE-7 for the purposes of adversarial or harmful user manipulation. We encourage caution when developing or training empathic models.

DATASET DETAILS

DATASET CONTENTS

SENSE-7 consists of 672 instances of multi-turn conversations with 4 different LLMs.

Each instance includes a Task and a list of Messages, with the following associated labels by the people that interacted with LLMs.

Tasks: Contains a list of conversational task and pre-/post-conversation metadata

- PID: Participant identifier
- Model: LLM version used for AI assistant
- TaskId: Unique conversation identifier
- PreTaskChoice: Pre-task conversation topic choice
- PreTaskImportance: Pre-task rating of importance of the conversation task (1 = Not at all important; 5 = Extremely important)
- PreTaskDesiredEmpathy: Pre-task rating of desired empathy (1 = Minimal empathy; 3 = High empathy)
- PostTaskExperience_Successful: Post-task rating of success in completing the task (1 = Strongly disagree; 5 = Strongly agree)
- PostTaskExperience_Engaged: Post-task rating of being engaged in the conversation (1 = Strongly disagree; 5 = Strongly agree)
- PostTaskExperience_PositiveInteraction: Post-task rating of having positive interaction with the AI (1 = Strongly disagree; 5 = Strongly agree)
- PostTaskExperience_UseAgain: Post-task rating of willingness to use the AI agent again for task (1 = Strongly disagree; 5 = Strongly agree)
- PostTaskEmpathy_*: Post-task ratings for 7 dimensions of empathy (1 = Strongly disagree; 5 = Strongly agree)

Messages: Contains conversation turns with empathy annotations

- MessageId: Order of the conversational turn
- TimeOffsetSeconds: Number of seconds since the start of the conversation (i.e., participant input)
- Role: user/assistant
- Content: Message text
- EmpathyOverall: Overall empathy rating per turn (1 = Very poor; 5 = Very good)
- 7 dimension ratings per turn: Affective, Cognitive, Response, Prosocial, Interest, Contextual, Relational (1 = Very poor; 5 = Very good)

The data was collected between February and May of 2024.

SENSE-7 does not contain links to external data sources.

DATA CREATION & PROCESSING

SENSE-7 was collected as part of a controlled user study by the members of the research team. Details about the study is fully described in our paper: <https://arxiv.org/abs/2509.16437>

The study involved 109 English-speaking information worker participants from a large technology company. From 695 conversations collected, the research team processed the raw data to remove all personal or identifiable details to arrive at 672 anonymized conversations.

PEOPLE & IDENTIFIERS

Data points in SENSE-7 correspond to individual people's natural conversations with LLMs, their self-reported ratings or information about the conversation topic, importance of the topic, desired level of empathy from the AI agent, per-turn rating of empathy, and perceptions about the conversation.

SENSE-7 does not include data pertaining to children.

The existing data that was used to create SENSE-7 did not contain information that could be used to directly identify a person.

Potentially identifiable information was manually removed from the existing dataset by researchers. Researchers reviewed each conversation to filter out sensitive conversations and systematically removing or replacing any personally identifiable and organization-specific information while maintaining the natural flow and meaning of the dialogue. Details about the redaction process is fully described in our paper: <https://arxiv.org/abs/2509.16437>

SENSE-7 is not believed to contain information that could be used to directly identify a person.

SENSITIVE OR HARMFUL CONTENT

The existing data that was used to create SENSE-7 contained information that might be considered sensitive or private, such as name, location, product or project information, and other confidential or demographic information that could be discussed in any naturalistic human-AI conversation.

To the best of our knowledge, the existing data that was used to create SENSE-7 did not contain information that might be considered offensive or insulting, or otherwise cause emotional distress.

Sensitive information was manually removed from the existing dataset by researchers. Researchers reviewed each conversation to filter out sensitive conversations and systematically removing or replacing any personally identifiable and organization-specific information while maintaining the natural flow and meaning of the dialogue. The information that was redacted consists of product or project information, and other confidential or demographic information. Details about the redaction process is fully described in our paper: <https://arxiv.org/abs/2509.16437>

SENSE-7 is not believed to contain information that might be considered sensitive or private.

SENSE-7 is not believed to contain information that might be considered offensive or insulting, or otherwise cause emotional distress.

OTHER PROCESSING

Details about the data processing steps is fully described in our paper: <https://arxiv.org/abs/2509.16437>

HOW TO GET STARTED

To begin using SENSE-7, please refer to instructions on <https://github.com/microsoft/sense-7/blob/main/README.md>

VALIDATION

To assess how effective SENSE-7 would be at its intended purpose, our team manually reviewed the dataset to ensure that the dataset reliably reflects diverse real-world communication scenarios. Our team also used the dataset to classify conversation empathy levels based on turn-level annotations and to evaluate predictive performance of the resulting model.

Specifically, we conducted exploratory modeling using a Large Language Model-based classifier to assess the feasibility of automatically detecting user-perceived empathy across five different levels. Evaluation showed encouraging accuracy and Spearman correlations, helping establish a starting baseline.

A detailed discussion of our validation methods and results can be found in our paper at:

<https://arxiv.org/abs/2509.16437>

LIMITATIONS

SENSE-7 was developed for research and experimental purposes. Further testing and validation are needed before considering its application in commercial or real-world scenarios.

SENSE-7 consists of English language instances only.

There are no instances of cultural and occupational diversity, non-English language use, multimodal empathy cues, and highly emotional or sensitive domains in the dataset. As a result, SENSE-7 is not expected to perform optimally at general-purpose, clinical/therapeutic, or cross-cultural empathic dialog modeling.

The ability to access external links in the dataset is beyond the control of the research team.

SENSE-7 has not been systematically evaluated for sociocultural/economic/demographic/linguistic bias.

Developers should consider the potential for bias as they select use cases, and evaluate and mitigate for accuracy, safety, and fairness concerns specific to each intended downstream use.

SENSE-7 should not be used in highly regulated domains where inaccurate or incomplete outputs could suggest actions that lead to injury or negatively impact an individual's legal, financial, or life opportunities.

BEST PRACTICES

It is the user's responsibility to ensure that the use of SENSE-7 complies with relevant data protection regulations and organizational guidelines.

The dataset does not include information about potential risks or harm from interactions. As empathic interactions can increase trust and engagement, we do not recommend using SENSE-7 for the purposes of adversarial or harmful user manipulation. We encourage caution when developing or training empathic models.

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ETHICS

Data collection and processing activities were conducted with approval from the Microsoft Institutional Review Board.

Data collection activities were conducted with the informed consent of the study participants.

Data contributors/ were compensated with a \$50 USD gift card. For each additional conversation, we compensated \$10 USD for a total compensation of up to \$100 USD per participant.

CONTACT

This research was conducted by members of [Microsoft Research](#). We welcome feedback and collaboration from our audience. If you have suggestions, questions, or observe unexpected/problematic data in our dataset, please contact us at sense7data@microsoft.com.

If the team receives reports of undesired content or identifies issues independently, we will update this repository with appropriate mitigations.